Machine Learning

14-05-26

GBM classifier

	x_1	x_2	y	Y	Z_0	$\hat{Y}_0 = \sigma(Z_0) = 1/1+e^{-Z_0}$
d_1			C_1	1	$\ln(\frac{4}{2}) = 0.69$	$1/1+e^{-0.69} = 0.66$
d_2			C_1	1	0.69	0.66
d_3			C_2	0	0.69	0.66
d_4			C_2	0	0.69	0.66
d_5			C_1	1	0.69	0.66
d_6			C_1	1	0.69	0.66

$C_1 \rightarrow +ve = 1$

$C_2 \rightarrow -ve = 0$

$$\text{odds of +ve class} = \frac{P(+ve)}{1-P(+ve)} = \frac{P(+ve)}{P(-ve)} = \frac{4/6}{2/6} = \frac{4}{2}$$

$$\log(\text{odds of +ve class}) = \log(4/2) = 0.69$$

↓

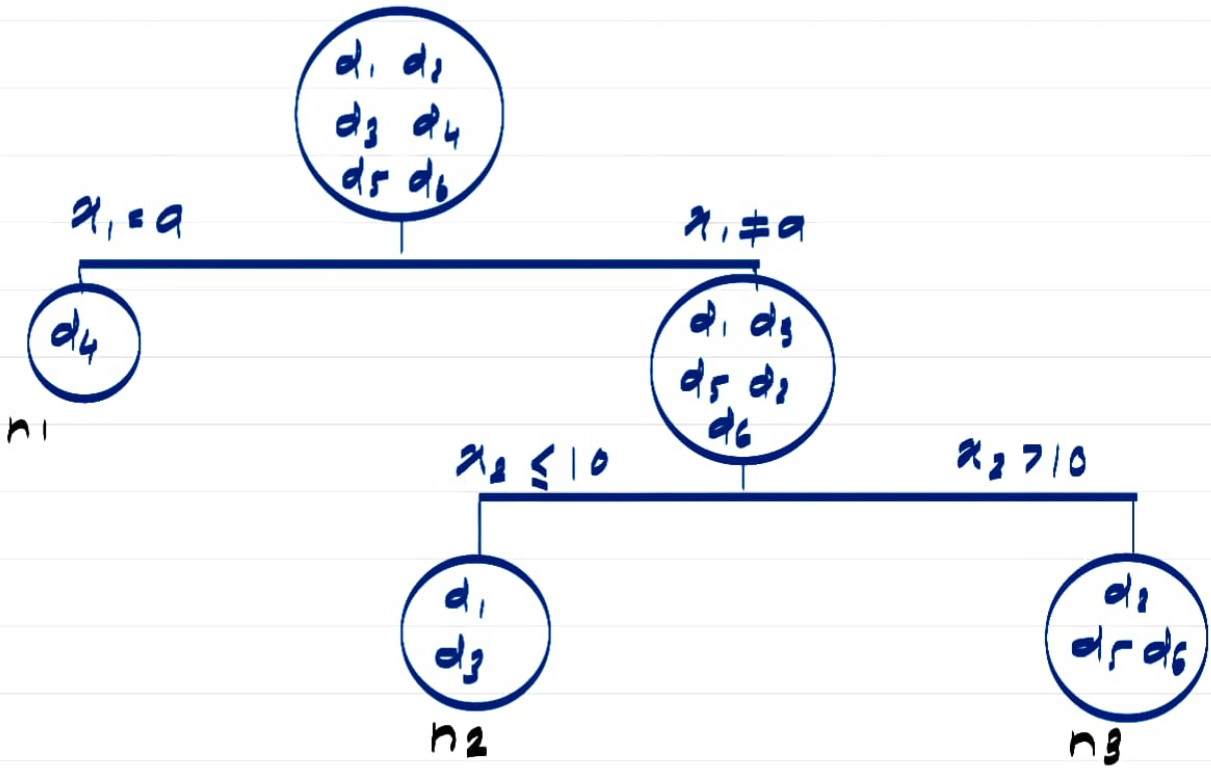
initial value of Z_0 before model training starts

	x_1	x_2	y	Y	Z_0	$\hat{Y}_0$	$R_1 = Y - \hat{Y}_0$
d_1			C_1	1	0.69	0.66	$1-0.66 = 0.34$
d_2			C_1	1	0.69	0.66	$1-0.66 = 0.34$
d_3			C_2	0	0.69	0.66	$0-0.66 = -0.66$
d_4			C_2	0	0.69	0.66	-0.66
d_5			C_1	1	0.69	0.66	0.34
d_6			C_1	1	0.69	0.66	0.34

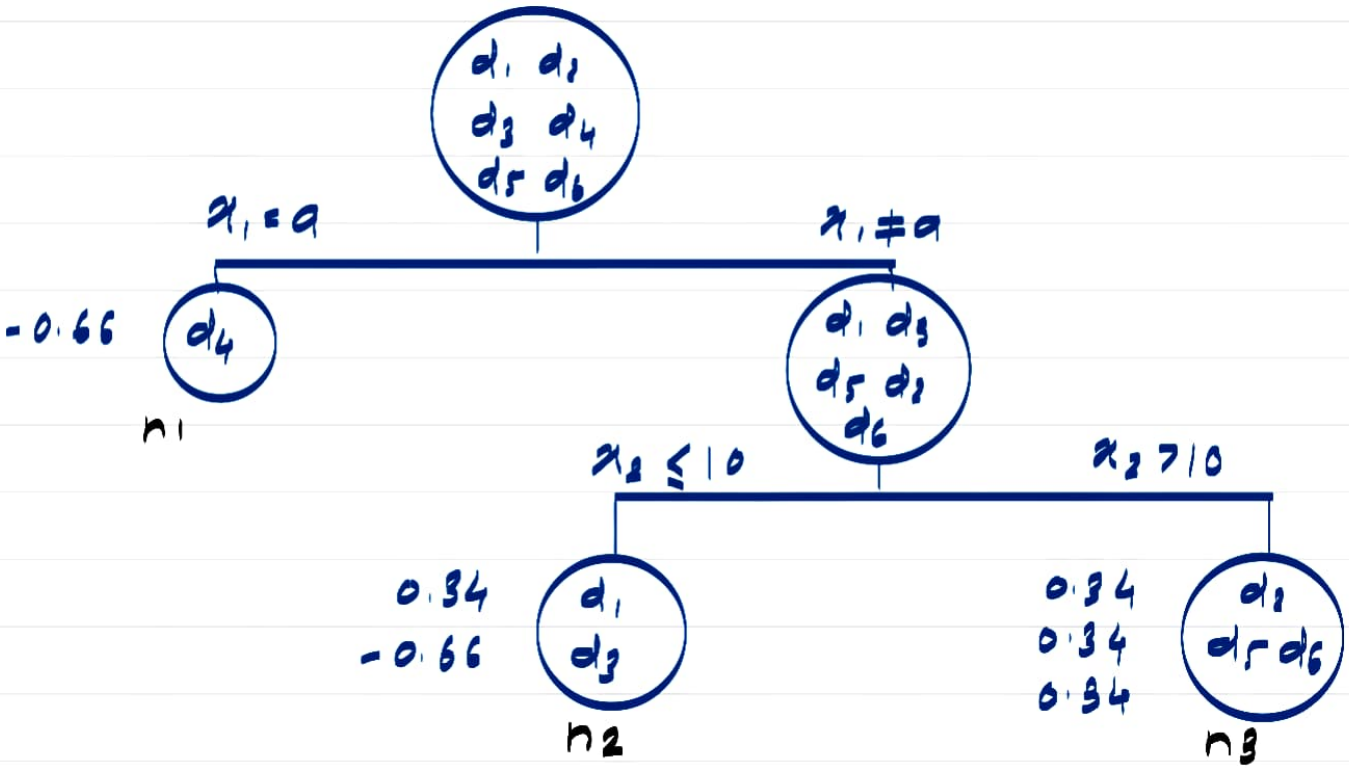
Now we train the model m_1 in the series with x_1 x_2 as features and R_1 as target

	x_1	x_2	y	γ	z_0	$\hat{\gamma_0}$	R_1	$\hat{R_1}$
d_1			c_1	1	0.69	0.66	0.34	
d_2			c_1	1	0.69	0.66	0.34	
d_3			c_2	0	0.69	0.66	-0.66	
d_4			c_2	0	0.69	0.66	-0.66	
d_5			c_1	1	0.69	0.66	0.34	
d_6			c_1	1	0.69	0.66	0.34	

In GBM classifier the base learner DTrees are regression models



output of node = $\dfrac{\sum \text{Residuals}}{\sum \text{prev prob} \times (1 - \text{prev prob})}$



$$\text{output of node} = \frac{\sum \text{Residuals}}{\sum \text{prev prob} \times (1 - \text{prev prob})}$$

$\hat{\gamma}_0$

$$\text{output of } n1 = \frac{-0.66}{0.66 \times (1-0.66)} = \frac{-1}{0.34} = -3$$

$$\text{output of } n2 = \frac{0.34 + (-0.66)}{0.66(1-0.66) + 0.66(1-0.66)} = -0.7$$

$$\text{output of } n3 = \frac{0.34 + 0.34 + 0.34}{0.66(1-0.66) + 0.66(1-0.66) + 0.66(1-0.66)} = 1.5$$

	x_1	x_2	γ	Y	z_0	$\hat{\gamma}_0$	R_1	$\hat{R}_1$	node
d_1			c_1	1	0.69	0.66	0.34	-0.7	n2
d_2			c_1	1	0.69	0.66	0.34	1.5	n3
d_3			c_2	0	0.69	0.66	-0.66	-0.7	n2
d_4			c_2	0	0.69	0.66	-0.66	-3	n1
d_5			c_1	1	0.69	0.66	0.34	1.5	n3
d_6			c_1	1	0.69	0.66	0.34	1.5	n3

$\alpha = 0.1$

	x_1	x_2	y	y	z_0	$\hat{y}_0$	R_1	$\hat{R}_1$	$z_1 = z_0 + \alpha \hat{R}_1$
d_1			c_1	1	0.69	0.66	0.34	-0.7	$0.69 + 0.1(-0.7) = 0.62$
d_2			c_1	1	0.69	0.66	0.34	1.5	$0.69 + 0.1(1.5) = 0.84$
d_3			c_2	0	0.69	0.66	-0.66	-0.7	$0.69 + 0.1(-0.7) = 0.62$
d_4			c_2	0	0.69	0.66	-0.66	-3	$0.69 + 0.1(-3) = 0.39$
d_5			c_1	1	0.69	0.66	0.34	1.5	$0.69 + 0.1(1.5) = 0.84$
d_6			c_1	1	0.69	0.66	0.34	1.5	$0.69 + 0.1(1.5) = 0.84$

$$\hat{y}_1 = \sigma(z_1)$$

	x_1	x_2	y	z_0	$\hat{y}_0 = \sigma(z_0)$	z_1	$\hat{y}_1 = \sigma(z_1)$
d_1			1	0.69	0.66	0.62	$1/1+e^{-0.62} = 0.65$
d_2			1	0.69	0.66	0.84	$1/1+e^{-0.84} = 0.7$
d_3			0	0.69	0.66	0.62	$1/1+e^{-0.62} = 0.65$
d_4			0	0.69	0.66	0.39	$1/1+e^{-0.39} = 0.6$
d_5			1	0.69	0.66	0.84	0.7
d_6			1	0.69	0.66	0.84	0.7

prediction after model $m_1 = \hat{y}_1$

Next steps :

compute $R_2 = y - \hat{y}_1$

train DTree m_2 with features x_1, x_2 and target R_2

↓

outputs $\hat{R}_2$ from m_2

↓

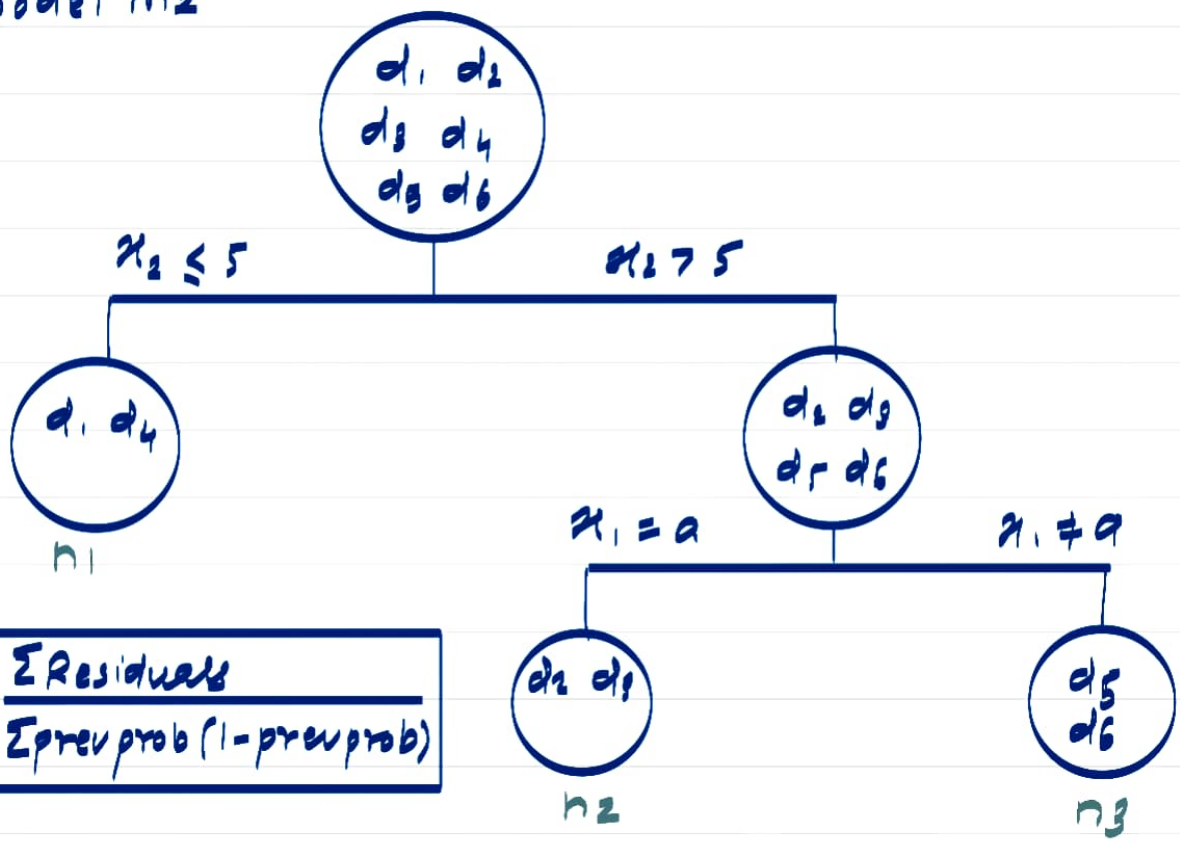
$z_2 = z_1 + \alpha \hat{R}_2$

↓

$\hat{y}_2 = \sigma(z_2)$ ← $R_3 = y - \hat{y}_2$

	x_1	x_2	y	$\hat{y}_0$	$\hat{y}_1$	$R_2 = y - \hat{y}_1$	$\hat{R}_2$	$z_2 = z_1 + \alpha \hat{R}_2$	$\hat{y}_2 = \sigma(z_2)$
d_1			1	0.66	0.65	$1 - 0.65 = 0.35$	v_1		
d_2			1	0.66	0.7	0.3	v_2		
d_3			0	0.66	0.65	0.35	v_2		
d_4			0	0.66	0.6	0.4	v_1		
d_5			1	0.66	0.7	0.3	v_3		
d_6			1	0.66	0.7	0.3	v_3		

model m2



output of node = $\dfrac{\Sigma \text{Residuals}}{\Sigma \text{prevprob}(1 - \text{prevprob})}$

output of $n_1 = \dfrac{0.35 + 0.4}{0.65(1-0.65) + 0.6(1-0.6)} = v_1$

↑ using $\hat{y}_1$ as prev. probabilities

output of $n_2 = \dfrac{0.3 + 0.35}{0.7(1-0.7) + 0.65(1-0.65)} = v_2$

output of $n_3 = \dfrac{0.3 + 0.3}{0.7(1-0.7) + 0.7(1-0.7)} = v_3$

Training GBM Classifier:

1) Decide the number of DTrees to build in series
   - decide all hyperparameters

2) start with initial prediction $\hat{Y}_0 = \sigma(z_0)$

$$z_0 = \ln(\text{odds})$$

3) compute residuals : $R_i = Y - \hat{Y}_{i-1}$

4) Train DT model m_i with features of original data and target R_i

5) compute residual predictions $\hat{R}_i$ from model m_i

$$\hat{R}_i = \frac{\sum \text{Residuals}}{\sum \text{prev. prob}\,(1 - \text{prev prob})}$$

6) compute $z_i = z_{i-1} + \alpha \hat{R}_i$

7) compute probability prediction

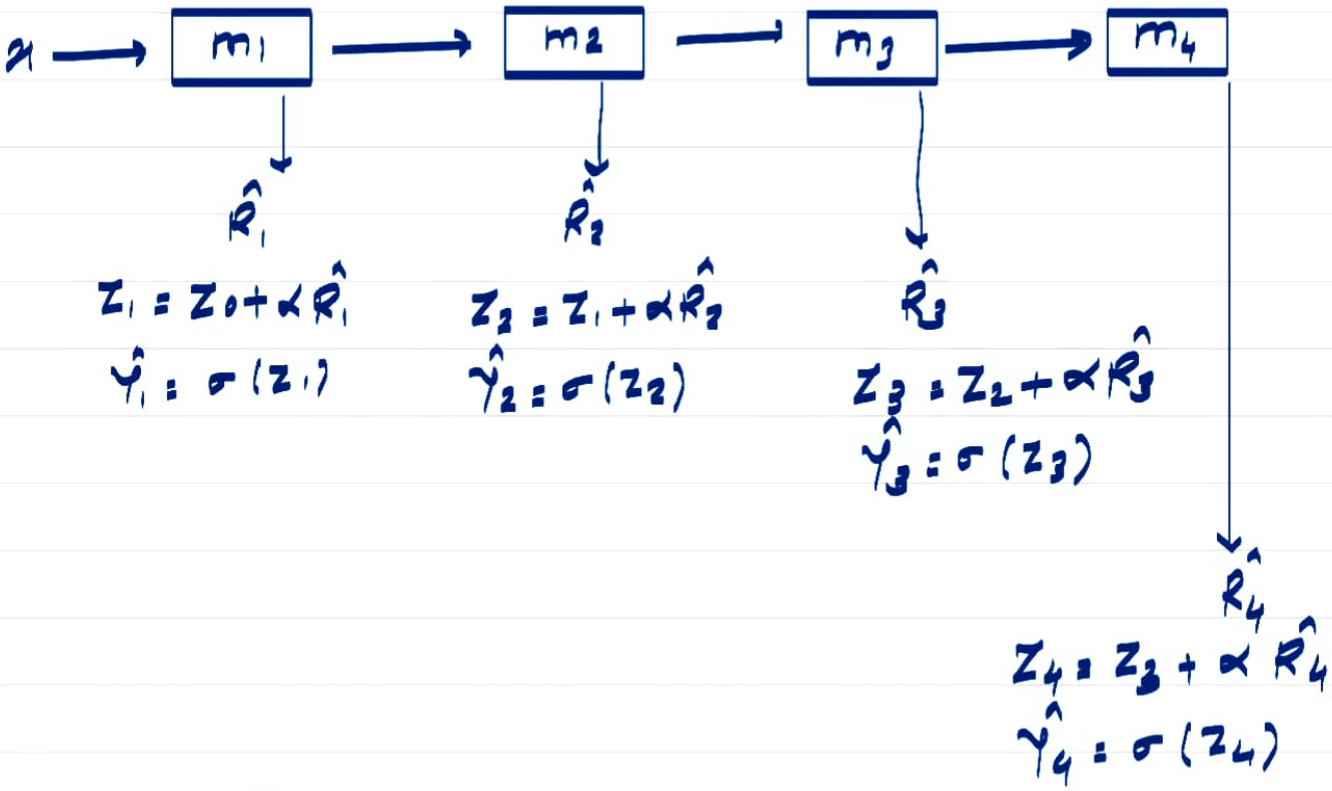
$$\hat{Y}_i = \sigma(z_i)$$

8) Repeat from step 3 to step 7 until stopping criteria is met

- stopping criteria is same as GBM regression

Inference of GBM

new datapoint



$\hat{Y}_4$ → final prediction of GBM classifier

eg.

$\hat{Y}_4 = 0.73$

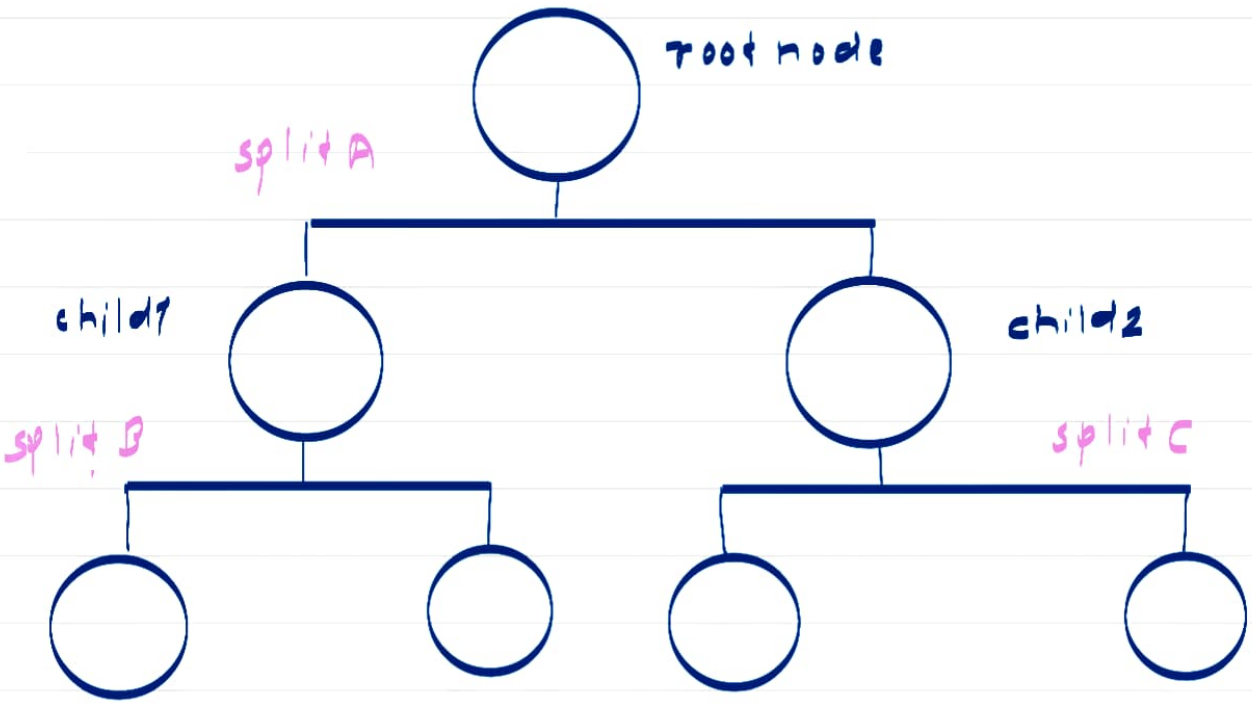
threshold = 0.6 for binary clf

$\hat{Y}_4 > 0.6$

predicted class = Y = +ve = C_1

XGBoost

- Extreme Gradient Boosting
- XGBoost adds improvements on top of GBM



GBM → split A → split B → split C — 9 sec
3 sec 3 sec 3 sec

XGB → split A → split B, split C — 6 sec
3 sec 3 sec 3 sec

- In GBM, DTrees are built in series as well as the splits within every tree are made in series
- In GBM, first all splits of left child are completed then splits of right child are completed
- This in series split of nodes makes GBM slow
- This is what XGB solves.